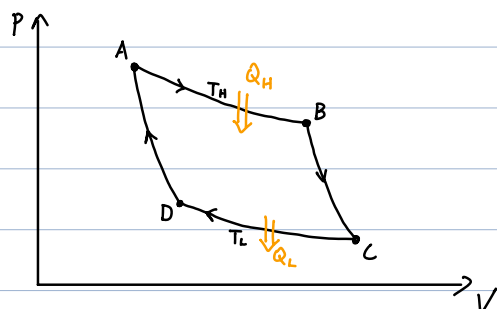


## The second law of thermodynamics

### Carnot cycle

1824, Sadi Carnot



$A \rightarrow B$  } Isothermal  
 $C \rightarrow D$  }

$B \rightarrow C$  } Adiabatic  
 $D \rightarrow A$  }

$A \rightarrow B$ : isothermal expansion

$$\Delta U = 0 \Rightarrow Q_H = W$$

$$= n R T_H \ln \frac{V_B}{V_A}$$

Similarly,  $Q_L = n R T_L \ln \frac{V_C}{V_D}$

$\Rightarrow$  Efficiency of Carnot cycle:

$$\begin{aligned} \epsilon_{\text{Carnot}} &= 1 - \frac{Q_L}{Q_H} \\ &= 1 - \frac{T_L}{T_H} \frac{\ln(V_C/V_D)}{\ln(V_B/V_A)} \end{aligned}$$

$D \rightarrow A$ : adiabatic

$$T_L V_D^{\gamma-1} = T_H V_A^{\gamma-1}$$

$$\Rightarrow \frac{T_L}{T_H} = \left( \frac{V_A}{V_D} \right)^{\gamma-1}$$

$B \rightarrow C$ : adiabatic

$$T_H V_B^{\gamma-1} = T_L V_C^{\gamma-1}$$

$$\Rightarrow \frac{T_L}{T_H} = \left( \frac{V_B}{V_C} \right)^{\gamma-1}$$

$$\left. \begin{aligned} &\Rightarrow \frac{V_A}{V_D} = \frac{V_B}{V_C} \\ &\Rightarrow \frac{V_C}{V_D} = \frac{V_B}{V_A} \end{aligned} \right\}$$

Finally,

$$\boxed{\epsilon_{\text{Carnot}} = 1 - \frac{T_L}{T_H}}$$

• Only depends on  $T_L/T_H$

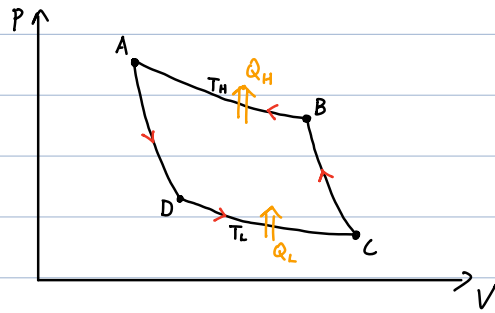
e.g.  $T_L = 303 \text{ K } (30^\circ\text{C})$ ,  $T_H = 503 \text{ K } (230^\circ\text{C})$

$$\Rightarrow \epsilon_{\text{Carnot}} \approx 40\%$$

e.g.  $T_L/T_H \sim 0$

$$\Rightarrow \epsilon_{\text{Carnot}} \sim 1$$

## Reverse Carnot cycle (Refrigerator)



- Work done **TO** the system
- Absorb heat **From** low-T reservoir; Release heat **TO** high-T reservoir

Coefficient of performance:

$$\begin{aligned} \epsilon^{(r)} &\equiv \frac{Q_L}{W} \\ &= \frac{Q_L}{Q_H - Q_L} \\ &= \frac{1}{1 - Q_L/Q_H} - 1 \\ &= \frac{1}{\epsilon} - 1 \end{aligned} \quad (r): \text{reverse}$$

For reverse Carnot cycle:

$$\begin{aligned} \epsilon_{\text{Carnot}}^{(r)} &= \frac{1}{1 - T_L/T_H} - 1 \\ &= \frac{T_L}{T_H - T_L} \end{aligned}$$

## Reversible process

Process where direction of System & Surroundings is reversed.

e.g. Frictionless quasi-static process

## Irreversible process

- e.g. - quasi-static process with friction
- adiabatic free expansion
- A cup of hot water
- all natural processes are irreversible

## The second law of thermodynamics

Clausius Statement :

(No perfect refrigerator)

Heat can never pass from a colder to a warmer body without some other change, connected with, occurring at the same time.

不可能把热量从低温物体传到高温物体而不引起其他变化。

Kelvin-Planck statement :

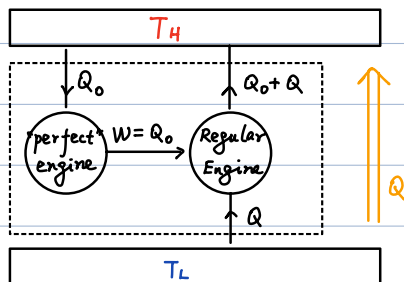
第二类永动机  
(No perfect heat engine)

It is impossible to construct a heat engine that, operating in a cycle, produces no effect other than the absorption of energy from a reservoir and the performance of an equal amount of work.

不可能从单一热源取热使它全部变成功而不引起其他变化。

### Equivalence between Clausius & Kelvin-Planck's statements

If Kelvin-Planck violated:

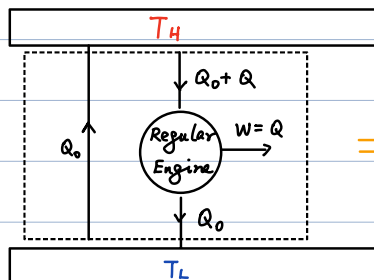


Total effect :

$Q$  from  $T_L$  to  $T_H$  without other effect.

$\Rightarrow$  Violation of Clausius statement.

If Clausius violated:



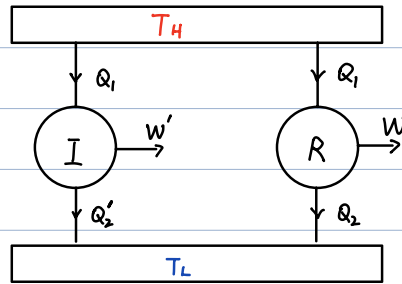
Total effect :

Convert  $Q$  from  $T_H$  to work without other effect.

## Carnot's theorem

- ① For heat engines working between two constant-T reservoirs, efficiencies of irreversible engines are always lower than reversible engines.
- ② For all reversible engines working between two constant-T reservoirs, the efficiencies are the same.

Proof:



$\left\{ \begin{array}{l} I: \text{Irreversible} \\ R: \text{Reversible} \end{array} \right.$

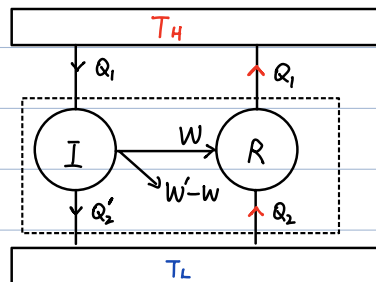
$$\left\{ \begin{array}{l} W' = Q_1 - Q'_2 \\ W = Q_1 - Q_2 \end{array} \right.$$

$$\epsilon_I = \frac{W'}{Q_1}, \quad \epsilon_R = \frac{W}{Q_1}$$

if  $\epsilon_I > \epsilon_R$  (i.e., if Carnot's theorem is wrong)

$$\Rightarrow W' > W$$

Design a combined process:



Net effect:  $\left\{ \begin{array}{l} \text{Absorb heat } Q_2 - Q'_2 \text{ from } T_L, \\ \text{Do work } W' - W = Q_2 - Q'_2 \end{array} \right.$

$\Rightarrow$  2nd Law violated

$$\Rightarrow \epsilon_I \leq \epsilon_R \quad \text{① proved}$$

Similarly, replace  $I \rightarrow R_1$ ,  $R \rightarrow R_2$

$$\Rightarrow \left\{ \begin{array}{l} \epsilon_{R_1} \leq \epsilon_{R_2} \\ \epsilon_{R_2} \leq \epsilon_{R_1} \end{array} \right.$$

$$\Rightarrow \epsilon_{R_1} = \epsilon_{R_2} \quad \text{② proved}$$

④

### Corollary of Carnot's theorem

For heat engines working between two constant- $T$  reservoirs, efficiencies of irreversible engines are always lower than Carnot engine.

$$\epsilon_I \leq \epsilon_{\text{Carnot}}$$